# Anthony Qi

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CS junior with experience building compilers, systems software, and backend-focused projects in C, Rust, and OCaml.

## EDUCATION

**University of Maryland, College Park** | *Expected Graduation: May 2028*
B.S. in Computer Science & Immersive Media Design (Computing)
**GPA**: 3.5/4.0 | **Award: Presidential Scholarship Recipient** (4 years)
**Relevant Coursework:** Systems Programming · Organization of Programming Languages · Algorithms · Discrete Structures · Object Oriented Programming · Linear Algebra · Data Structures and Algorithms

## TECHNICAL SKILLS

**Languages:** Java, C, OCaml, Rust, Python, JavaScript, C#, SQL, x86-64 Assembly

**Other Skills:** Git, Bash, Linux/Unix, GDB, Valgrind, JUnit, MySQL

**Concepts**: Compilers, Systems Programming, Multithreading, Memory Management, Garbage Collection, Static Type Systems, Data Structures & Algorithms, Machine Learning, NLP Model Training (Intent Classification)

## PROJECTS

**Personal Portfolio Website (React, Typescript, Vite) |** *Summer 2026*
- Rebuilt personal portfolio site from vanilla HTML/JS/CSS into a React, Typescript, Vite, and Tailwind CSS deployed via Github Actions to Github Pages.

**Rust Garbage Collection (Rust) |** *Spring 2026*
- Implemented three garbage collectors in Rust: reference counting, mark-and-sweep, and stop-and-copy, including recursive heap traversal and live-object compaction.

**OCaml C Compiler (OCaml) |** *Spring 2026*
- Built a full compiler pipeline in OCaml: lexer, parser, AST evaluator, static type checker, and stack-machine code generator for a SmallC-like language.

**Unix Shell (C) |** *Fall 2025*
- Built a Unix-style shell from scratch supporting fork/exec, I/O redirection, pipelines, and background processes via wait() and signal handling.

**Explicit List (EL) Malloc and Matrix Operation Optimization (C) |** *Fall 2025*
- Implemented an explicit free list allocator using mmap, with splitting and coalescing for dynamic heap management.
- Optimized multithreaded matrix multiplication using pthreads, mutex synchronization, and cache-aware blocking to improve memory locality and performance.

**Minecraft Fabric Mods (Java) |** *Jun. – Sep. 2025*
- Built two Minecraft Fabric mods featuring server-side state tracking (boss bar + commands) and custom gameplay mechanics (enchantments, cooldown systems, and textured items).

## EXPERIENCE

**MascotGo | Remote |** *Jan. 2026 – Present | Student In Residence*
- Fine-tuning a lightweight NLP model using Unsloth on a dataset of 11,000+ labeled queries for intent classification across student support categories.
- Collaborating with a cross-functional team to clean and standardize training data, validate model accuracy, and route classified queries to appropriate response tiers.

**National High School Game Academy | Carnegie Mellon University |** *Jul. – Aug. 2023 | Student Game Developer*
- Built a fully playable game prototype in Unity/C# with a multidisciplinary team, integrating original art and audio under sprint-based deadlines.
- Presented the final game to CMU faculty and industry professionals at a public showcase.

**ROAR Club | Valley Christian High School |** *Jan. 2023 – Jun. 2024 | Co-Founder & VP*
- Co-founded a machine learning club focused on autonomous racing; taught ML concepts through presentations to 20+ members.

**Home Depot | San Jose |** *Jun. 2025 – Aug. 2025 | Service Specialist*
- Handled customer inquiries, returns, and equipment rentals (trucks, vans, carpet cleaners) in a high-volume retail environment.

## ADDITIONAL

**Languages:** English (native), Chinese (fluent), Japanese (reading), Spanish (basic) | **Creative:** Drawing, 3D Modeling, Digital Media